FIESTA: Comprehensive Analysis Handbook

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Background

FIESTA is a single molecule motility analysis tool. Input multi channel tif movies and does automated particle tracking using gaussian fits, track processing, and motility parameter estimation and other related statistics all packaged in an easy to use gui with many saveable data spots along the way.

This document is organized in the way:

1. Installing this software

2. FIESTA Basics - the basic demo tutorial
3. Navigation
4. FIESTA for General Motility Assays
5. FIESTA for Stepping Analysis

~~FIESTA homepage:~~
~~https://fusionforge.zih.tu-dresden.de/plugins/mediawiki/wiki/fiesta/index.php/FIESTA~~

1. Installing FIESTA

1.1. System Requirements:

- For MATLAB r2021a, use this code version with the old FIONAviewer
- The most up to date version runs on r2025a, use this code version
  - MATLAB student licenses are free through the university. Look at the instructions at this link:
- Windows OS and Mac OS both work
- MATLAB Dependencies:
  - Image Processing Toolbox
  - Optimization Toolbox
  - Signal Processing Toolbox
  - Statistics and Machine Learning Toolbox
  - Parallel Computing Toolbox (optional, Windows only)

1.2. Download from Repository

~~https://fusionforge.zih.tu-dresden.de/plugins/mediawiki/wiki/fiesta/index.php/Documentation#Installing_FIESTA~~
The most recent version can be downloaded in a zip file from Yildiz Lab GitHub (https://github.com/Yildiz-Lab/YFIESTA) or by using git clone command which is more efficient for updating (for how to use git, check out this shared document https://docs.google.com/document/d/1AZxG3sWQdwlwqyjVYWb_WwkCwdv46Kax356xoW2YbHc/edit#heading=h.r85at0frstn9)

- For MATLAB r2021a, use this code version with the old FIONAviewer
- The most up to date version runs on r2025a, use this code version

1.3. Run Fiesta in MATLAB terminal

Open a MATLAB r2021a terminal (or another version will work probably) and navigate to the folder where your Fiesta source code is located. Then run the following command: Fiesta

A few moments later, a gui should pop up

2. FIESTA Basics

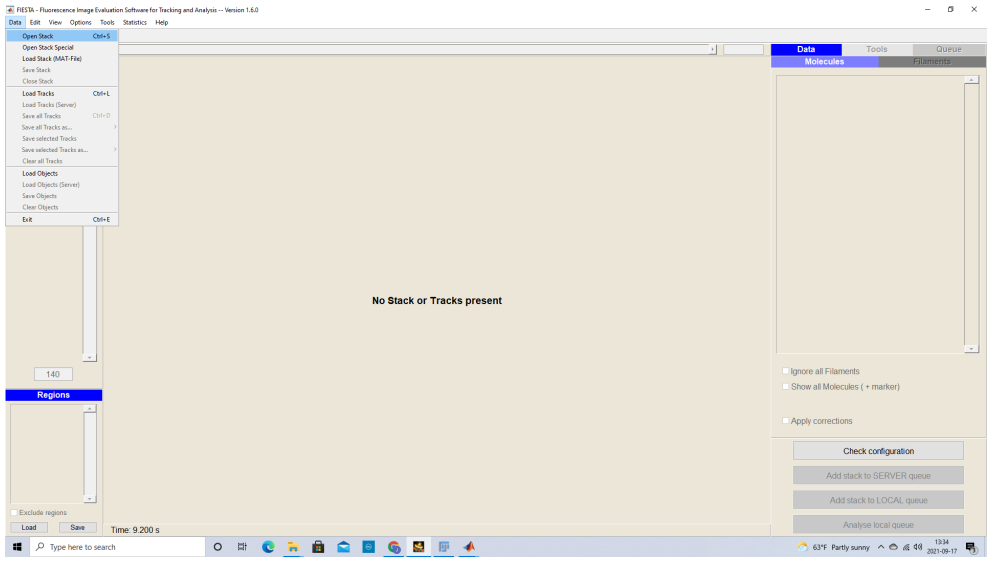
Note that all screenshots are done with the demo data which you should be able to follow along in the provided Demo folder in the shared FIESTA package on the Yildiz Lab Drive

2.1. Opening a File in FIESTA

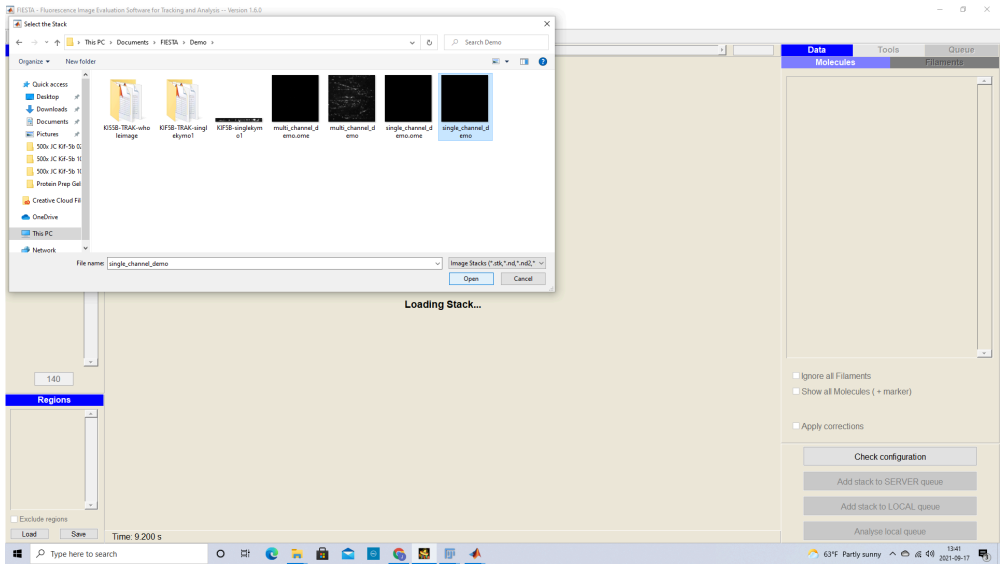
FIESTA naturally opens .tif stacks. Depending on whether your tif is dual movie or not may affect how you open it

- Single channel

Navigate to Data > Open Stack



Select your file you want to process then press open



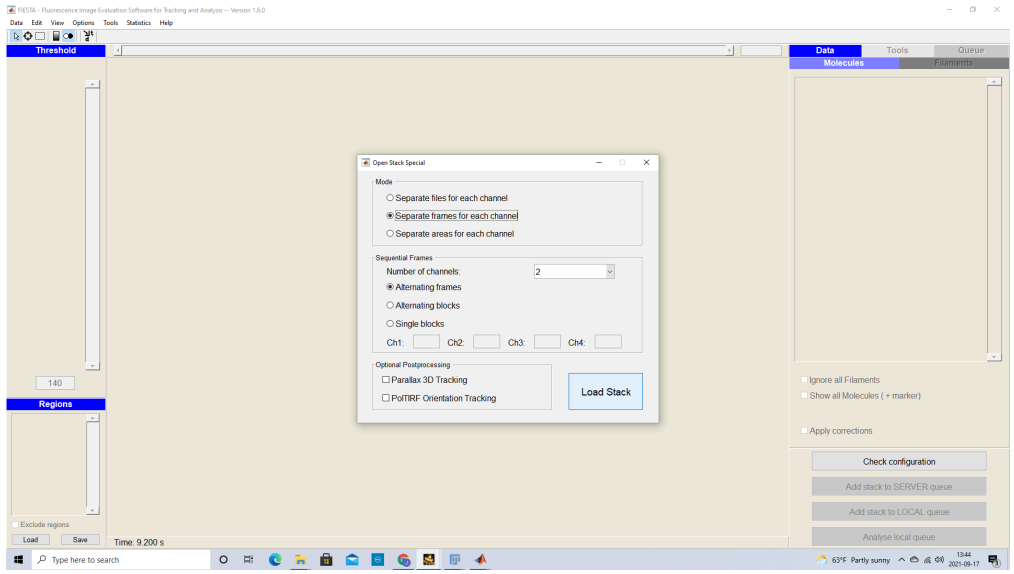
There will be a **smoothing sigma** option presented. This is an option related to MATLAB's gaussian filter function which overall smooths the data. This sometimes makes it easier for MATLAB to identify particles later. The default **0** is the original image.

There will be an automatic code that loads the image and saves it as a .tif that is readable by FIESTA. This will be the same string but with a .f to recognize it as a FIESTA readable tif file. It will always attempt to load this in the future unless you change the .f addition.

Enter the time between frames (ms) and pixel size (nm) respectively.

Multi-channel

Navigate to **Data > Stack Special** and you should open the following menu:



For time splitting (such as the b307 scope way of imaging), you should enter the **Separate Frames for Each Channel** and then the number of channels before clicking **Load Stack**.

Then select the muti-channel movie and enter the parameters for movie time and pixel size.

Note that you will be asked for the movie time twice. The movie time entered should be the total time from one frame of the same channel to the next. For example: if exposure time is 250ms for each (like in the demo), then enter in 500ms for each channel to get correct velocities and distances

- Folders

  By selecting the **Data > Open Folder as Stack** option, you now have the ability to select an entire folder and it will load all the movies into a single .tif file. This is useful if you only care about one frame tif movies and just want to track objects efficiently.

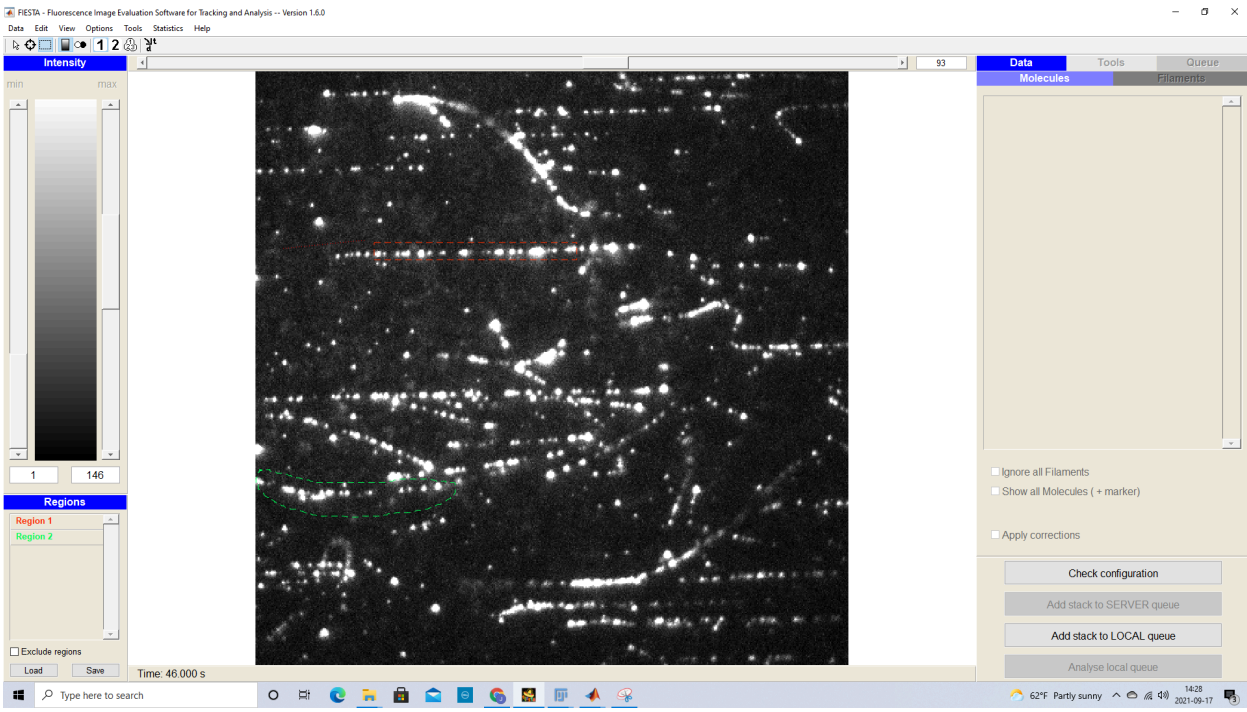
2.2. Selecting Regions (or not)

Use the region selecting tool to select multiple regions to analyze or to exclude multiple regions from being analyzed (i.e. moving MTs, etc). Simply use the Region Selection Toolbar and highlight the regions you want which should be color coordinated and added to the Regions List Panel. Right click on regions in the list to modify or delete from selection.

Note that regions are automatically applied to all stacks added to the queue (unless of course the exclude regions checkbox is toggled, then everything but the regions are added).

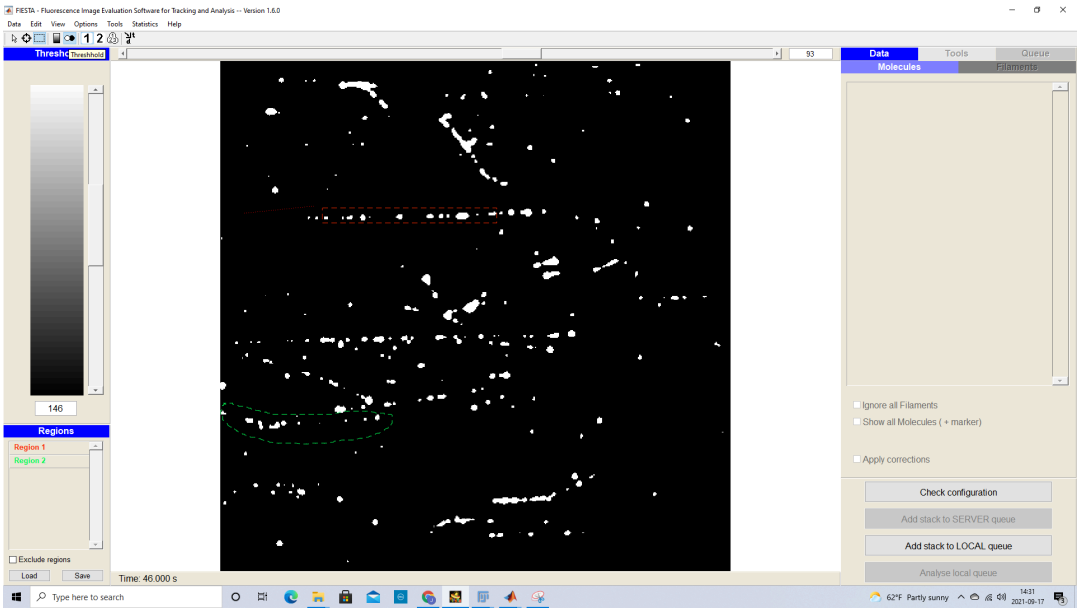
If you want to add Scan Region elements more like how a kymograph is traced out, check out Sec 3.1. For **automated scan region selection**, look at Sec 3.1.1.

If you do not select any regions, the code will just assume it should analyze the whole movie.



2.3. Set Appropriate Threshold

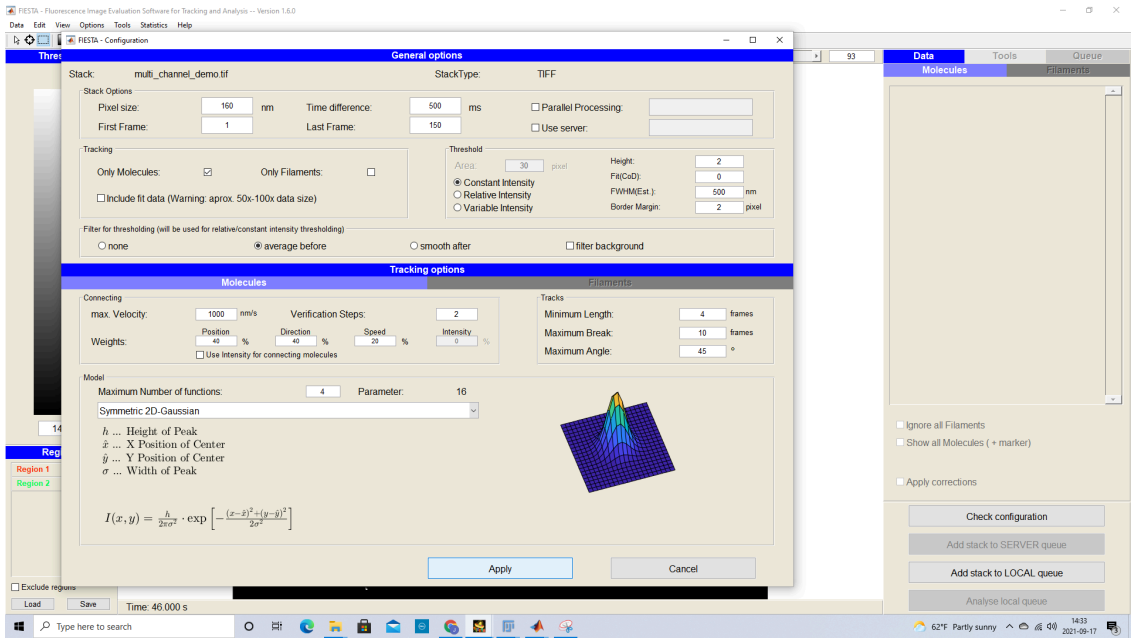
FIESTA by default uses the **Threshold** application to find objects. This is an independent view from the **Intensity View**. Therefore, make sure to toggle the white and black circle in the **View Mode Toolbar** and set an appropriate threshold. There is more clarification on what is appropriate in the documentation: https://fusionforge.zih.tu-dresden.de/plugins/mediawiki/wiki/fiesta/index.php/How_to_set_a_good_threshold



2.4. Configuration

Use the right analysis panel now that everything is set and check that your configuration is what you want. For the most part, this can be left default unless something in the code is really going wrong. However, it may be wise to select the “Only Molecules” option since we only want to see particles and not worry about filaments. This will speed up the analysis process.

Make sure to press apply at the end to save changes.

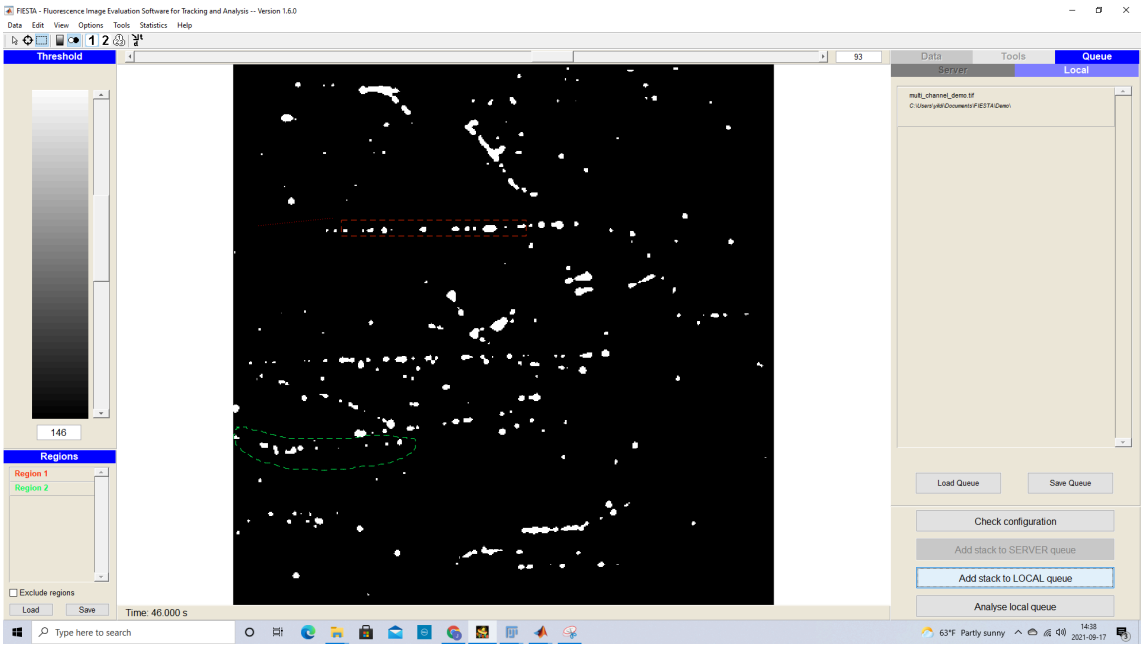


General advice for choosing the Configuration:

- THE DEFAULTS ARE DEFAULT FOR A REASON! THEY ARE A GOOD PLACE TO START!!!
- If analyzing stepping or movies where particles are less dense on MTs, have a higher weight in position for connecting tracks.
- If analyzing crowded movies or where molecules move consistently fast, have a higher weight for direction
- Set max velocity to about 4x the average velocity you expect. It doesn't usually hurt to have it higher.
- Maximum Angle can usually be set to a smaller value for straight MTs
- Choose an appropriate Maximum Break for blinking and connecting crossing tracks.

2.5. Add Stack(s) to LOCAL queue

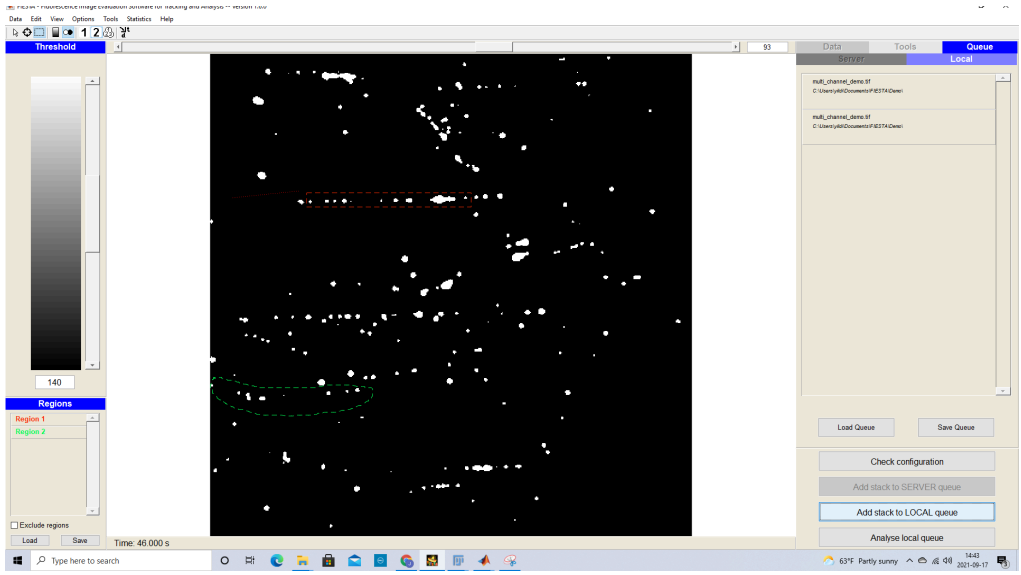
Simply press **Add stack to LOCAL queue** (or *Hotkey y*) and the **Queue Tab** should now have a .tif file listed as shown.



IMPORTANT NOTE: FIESTA may give a warning in the command line claiming that there > 100 molecules in the stack. One can simply add anyway if desired. ~200 molecules can be analyzed in 10 minutes. However, if you have a bad threshold or have a lot of noise, you may want to be careful since the analysis will lead to false molecules.

- Multi-channel

  For multi-channel use, make sure to switch your stack and add this other stack separately. You can toggle between stacks using the Channel View Toolbar. Both should show up in queue and will do one after the other. It will also keep track of which channel is which when you load it.

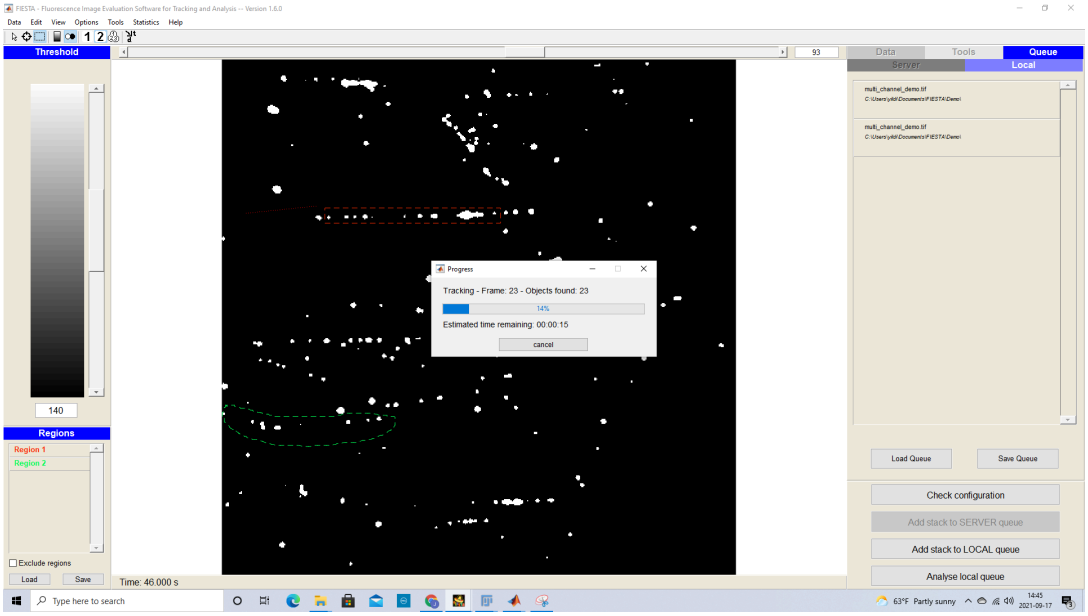


2.6. Analyse Local Queue

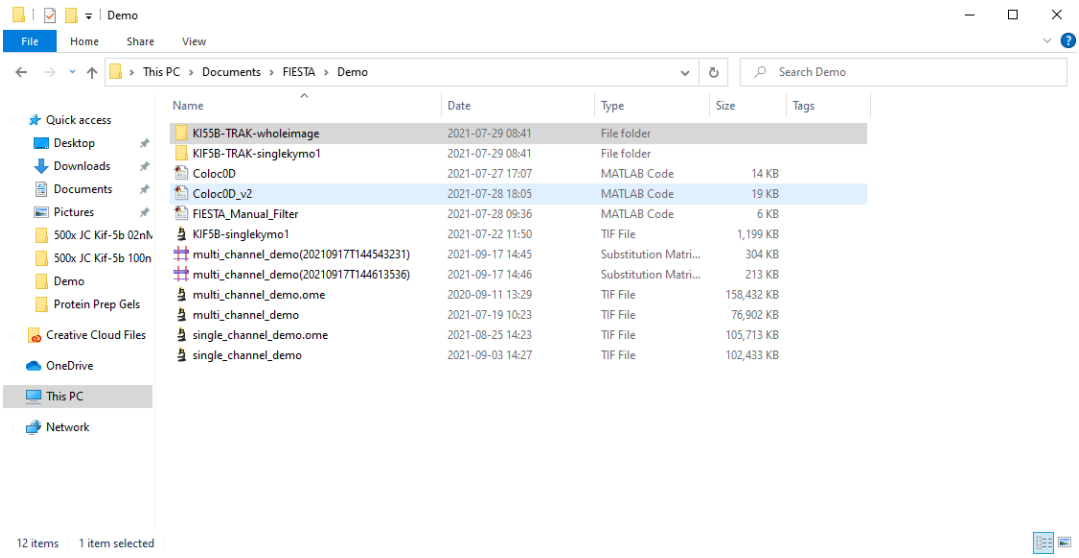
Press “Analyse local queue” and watch it count down. Check email, go take a look at some more movies, check on your protein prep, take more images, get lunch (if you have enough time), watch a fun video, paint miniatures, or just watch paint dry while twiddling your thumbs.

It will find all objects, then process tracks, then postprocess tracks.

WARNING: Once you press this button, FIESTA will be unresponsive until it finishes.



Once the post-processing finishes, the data should all be saved to a matrix substitution file which can be loaded into MATLAB or read by FIESTA directly. It is saved in the same directory as your stack and named by default: *stack name passed* + (*date, time, seed*)

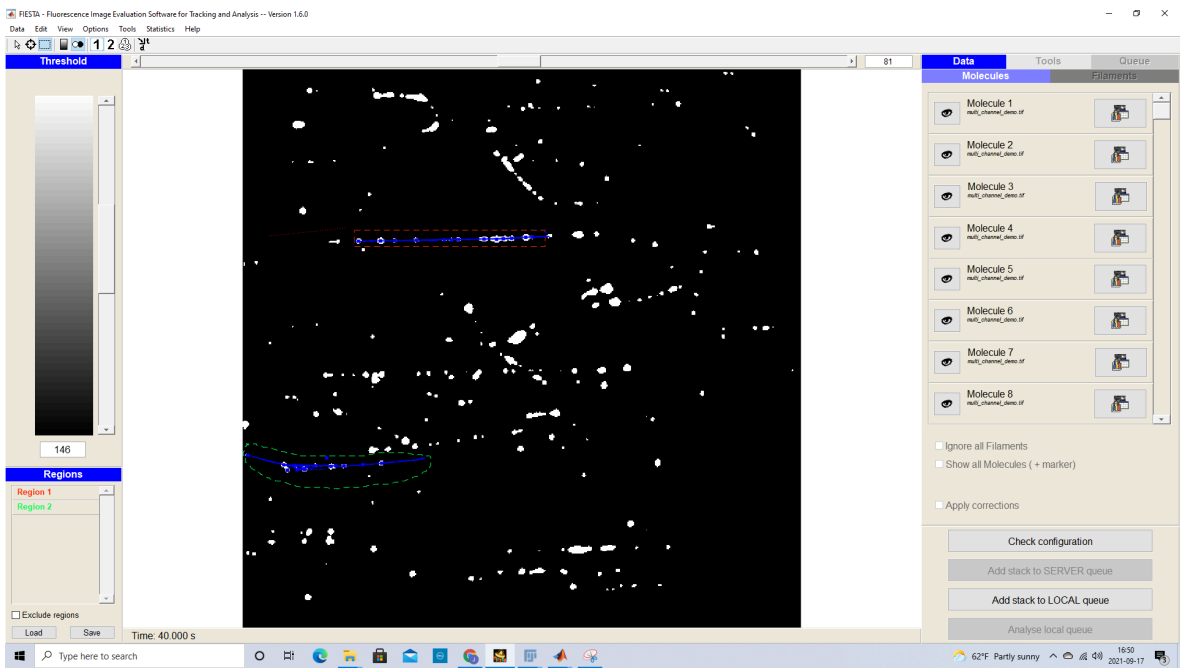


2.7. Load Tracks

All tracks can be loaded in using the Data > Load Tracks tool. Then select the Sub Matrix file output by your “Analyse Local Queue” step.

You should now see blue lines trace out paths of molecules and blue dots following the centers of molecules.

In addition, your Data > Molecules Panel on the right side should now have a list of molecules associated with it.



2.8. Filter/Select Tracks

FIESTA strives to pick up any bright spots as molecules. However, this also means it picks up many molecules that we don't consider walkers (too small, no movement, etc.). Therefore, we can filter our results, either using a built in filter or a manual filter constructed in MATLAB (see 2.7).

- Select All Tracks in list

  Simply right click anywhere in the Data > Molecules panel and select "Select all Molecules" from the options

- Select only Moving Objects using the built in filter

  Under the Edit tab, there is an option to “Find All Moving Objects”. Upon selecting this, you will determine the minimum travel distance and frame lifetime in order to be counted as a selected molecule. Simply type these in and press enter.

Selected molecules will be highlighted blue in the Data > Molecules panel

2.9. Optional: Manual Post Processing

There are many manual edit options available if wanting to fine comb your data. At some level, this becomes tedious and not incredibly beneficial. However, if desired, FIESTA offers a multitude of manual processing options once the tracks are loaded.

- Individual Chart / Path changing

  Selecting the table icon next to any molecule pulls up a large window with a multitude of editing options. Changing anything in here will change the molecule in the window. However, don’t forget to save your data if you do anything you want to keep for later.

  Some things you can do:
  - Change Color of certain tracks for identification
  - Apply tags to identify certain molecules compared to others
  - Change plot view to see more characteristics of the molecule
  - Select sections of data to remove

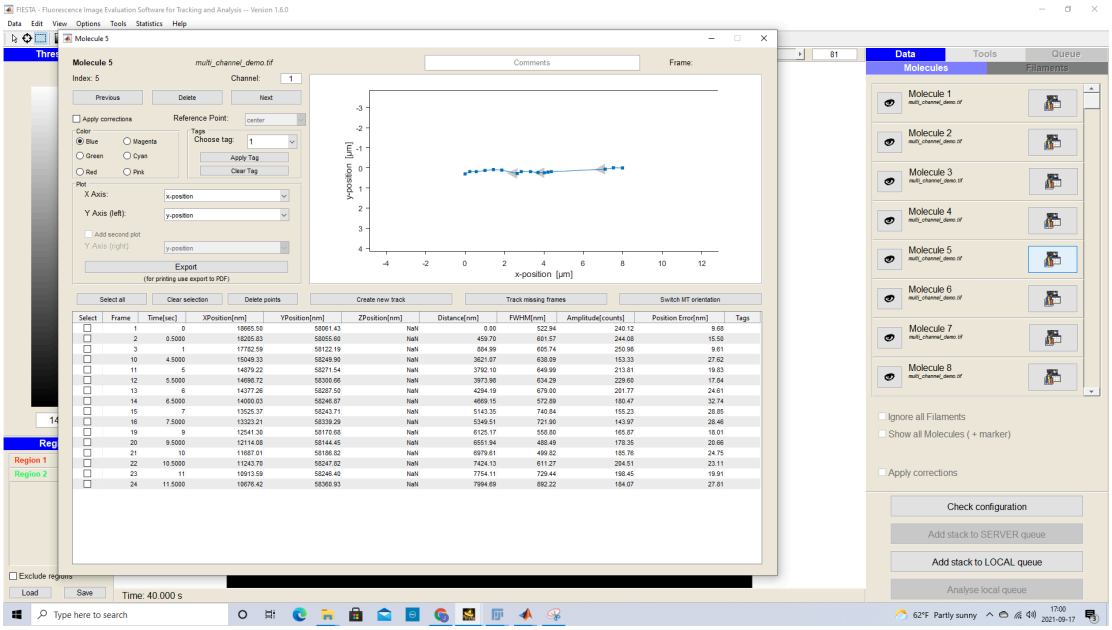


Image 2.1 Molecule Chart Panel

- Joining Selected Tracks

  In selecting multiple tracks, you can join them by right clicking on the Data > Molecule panel and join selected tracks

2.10. Get Statistics and save them for later

Warning: Statistics will only be calculated for molecules selected (highlighted in blue on the Data panel and/or selected on the actual image)

Error: If both molecules and filaments exist, there might be an error message that comes from trying to calculate Statistics that claim “Both Molecules and Filaments cannot be included”. Simply toggle on/off one cycle the “Ignore all Filaments” checkbox and now it should work.

All statistics related calculations are in the main tab “Statistics”. A brief overview of some of the important ones to use:

- Path Statistics - auto calculates a fit path and returns MSD and other measurements
- Velocity Statistics - compiles a total histogram of instantaneous velocities with the capability to fit gaussians

- Estimate Motility Parameters - uses bootstrapping methods to estimate velocity, distance, and interaction time of molecules.
    - Warning: this method requires at least 7 molecules to work. Otherwise it will just get stuck or never start.
-

~~2.11. (Not Completely Functional) Automatic Titration Curve Plotting~~

~~Initial Template in the following sheet and colab notebook:~~

~~Google sheet with recorded data from FIESTA analysis:~~
~~https://docs.google.com/spreadsheets/d/1ykWOhJPE7GB2lrw6GjHoUvRucqWnjQhA8yNAMjJsabE/edit?usp=sharing~~

~~Google Colab Notebook for plots (using data from spreadsheet):~~
~~https://colab.research.google.com/drive/1C3EstDEWcUhuTYTYBtTLEuLIPuCrhotO?usp=sharing~~

3. FIESTA for General Motility Assays

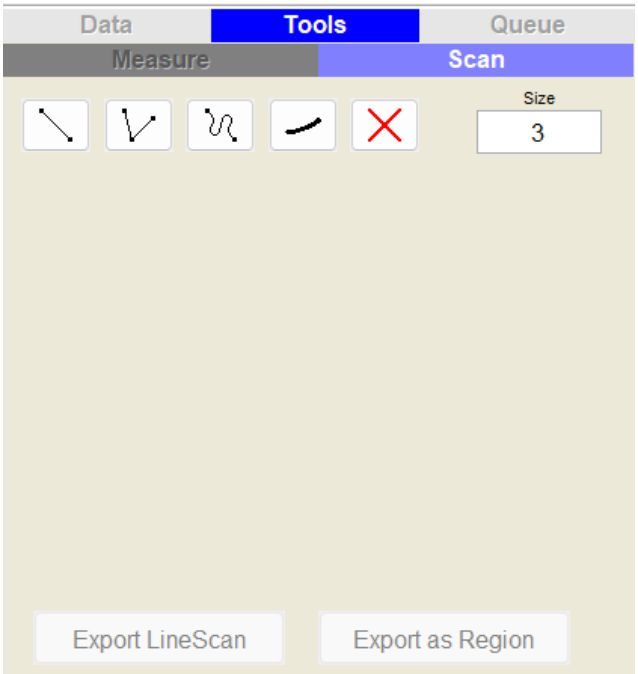
We have optimized some conditions to do general motility assays. Make sure you are in branch “main” of the YFIESTA package. You can verify this by using the command: git checkout main

Windows PowerShell

```
PS C:\Users\yildi\Documents\GitHub\YFIESTA> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\yildi\Documents\GitHub\YFIESTA> git checkout main
Already on 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\yildi\Documents\GitHub\YFIESTA>
```

3.1. Choosing Regions with Scan

To more appropriately align with Yildiz Lab general motility, we want to keep track of individual MTs/filaments. In order to do this, one can use the builtin Segmented Line Scan Tool located in the Right Panel > Tools > Scan Option.



Once drawn, you can export this as region which will save the area as a scan region with important parameters to select later.

One may find it easier to do this with the View > Z Project in the toolbar

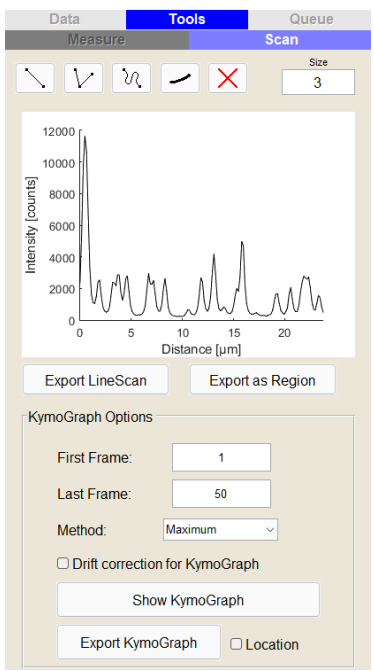
■ (Beta) Automated Region Finding

Developments have been made to try and automatically find regions for kymograph analysis. For more information on doing this, check out the ASSAMM project in more detail development notes.

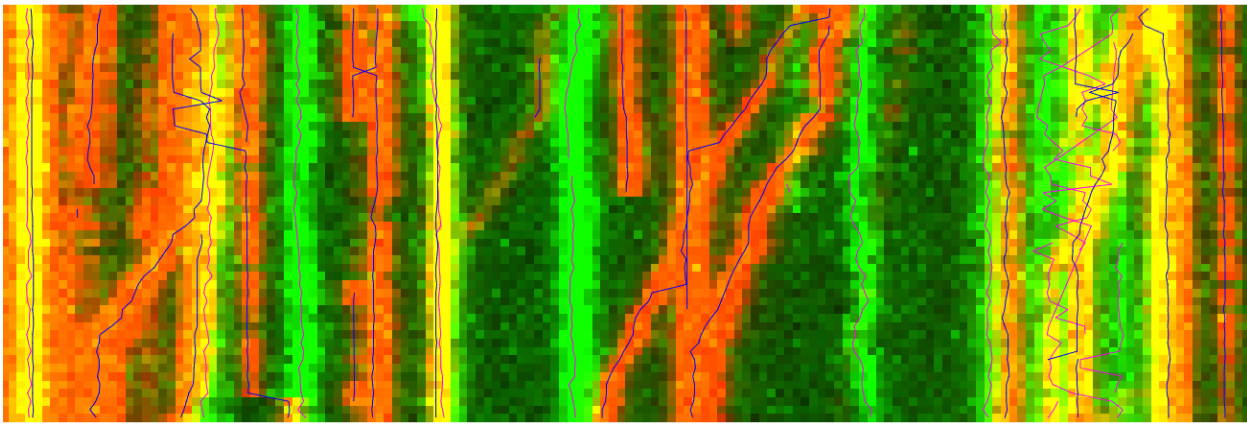
For the time being, you can just use the “Find Regions” button in the left panel while on whichever channel shows the cleanest Z-projection and it will spit out some suggested regions to analyze. Note that if you want thicker regions, you must select the Scan Size parameter in Tools > Scan

3.2. Analyze Kymographs

After adding to queue, analyzing, and loading in tracks as described before in Section 2.4-2.7, you can now load in a specific selected filament by selecting the specific scan you want to load. The panel in the upper right should update with the new scan information.



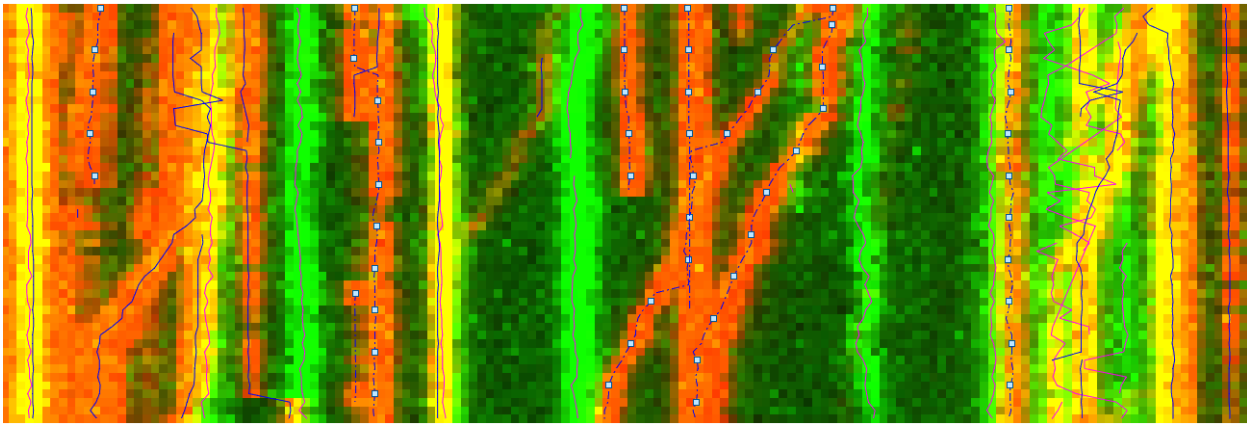
By selecting Show Kymograph, a Kymograph should then be displayed in the main mid Panel.



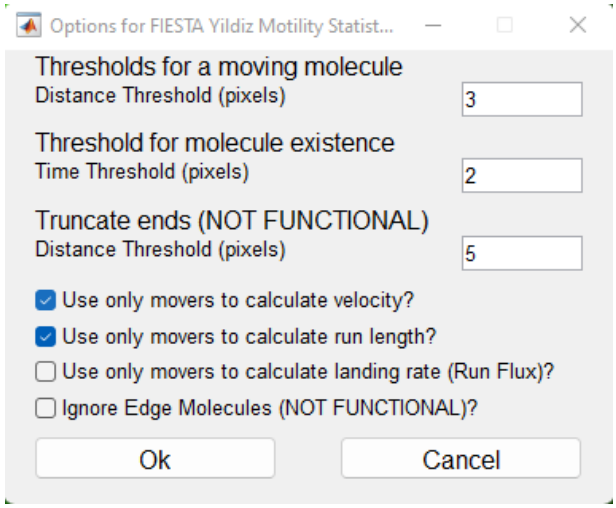
In this image, we selected all channel 1 molecules to be blue and channel 2 to be magenta. This can be done easily according to Filer/Select Tracks in Section 2.8

3.3. Motility Parameters

After selecting molecules one wants to analyze, your image should look something like below. (It might be easier if only do one channel at a time)



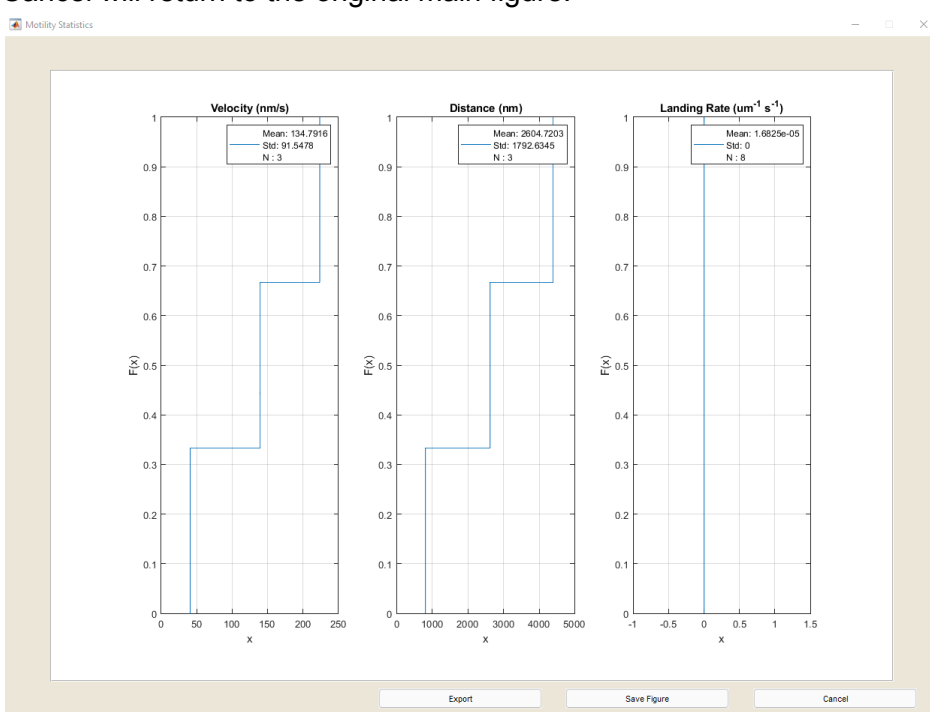
Then, going to Statistics > Yildiz Motility Parameters will prompt another dialog box



All these options are editable for the user to change how the statistics of the selected molecules are calculated. Then press Ok.

3.4. Evaluate and Export Motility Parameters

After selection, on pressing Ok, the results will show up in a cdf distribution. You can export these to an excel spreadsheet that will append all of the statistics from the same file entitled the same as the main file.
Simply press Export to save the data to the excel spreadsheet.
Save Figure will allow for the current figure to be saved.
Cancel will return to the original main figure.



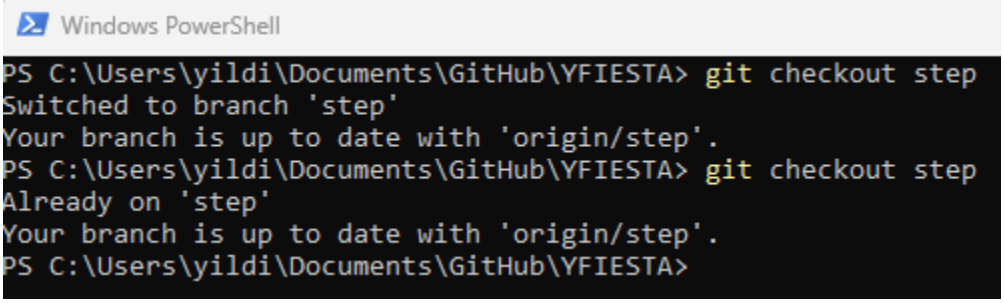
3.5.

4. FIESTA for Stepping Analysis

FIESTA has the capability to track single molecules of any speed, but it shines very well in cases of dilute, bright molecules. Stepping assays are intentionally at this regime so FIESTA has been extensively developed to handle these cases.

There is a separate stepping branch of YFIESTA,mostly for the case of user defined tracking parameters, but it is still useful to involve less user input. If you intend to use YFIESTA for stepping, switch to branch step using the git checkout command.

Note that Sec 4.2-4.5 are very similar to generic motility assays. If you are comfortable with this selection, it is pertinent to skip to Sec 4.6.



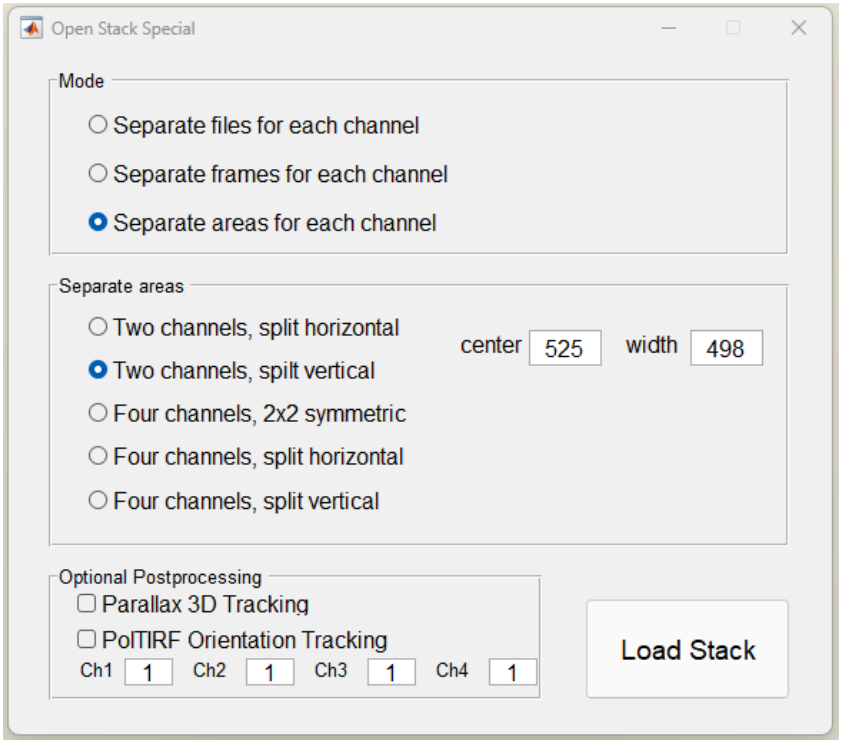
4.1. Open Stack Special (for two color movies)

The acquisition of data on the stepping scope (or anything with a emission splitting Gemini-type piece of optics) utilizes spatial splitting. In order to load in your data in an overlay view, use the **Separate Areas for Each Channel** option.

If you want to adjust which pixel the center is split and the width/height of each channel, you can change those parameters on this GUI also.

Finally, there exists an option to sum n successive frames together by channel and set them all equal. This is a post-processing option that could help with decreasing background noise. The default is to do 1 successive frame, keeping all frames unique. However, if you wish to set n ≠ 1 for any channel, you can do this in the Optional PostProcessing dialog box.

Once you press **Load Stack** then you will be prompted to select the movie you want to load as was outlined in Sec 2.1. Unlike the multichannel there, however, make sure that the time difference is the actual acquisition time.
I.e. 100 ms experimental exposure, each time between frames should be 100 ms

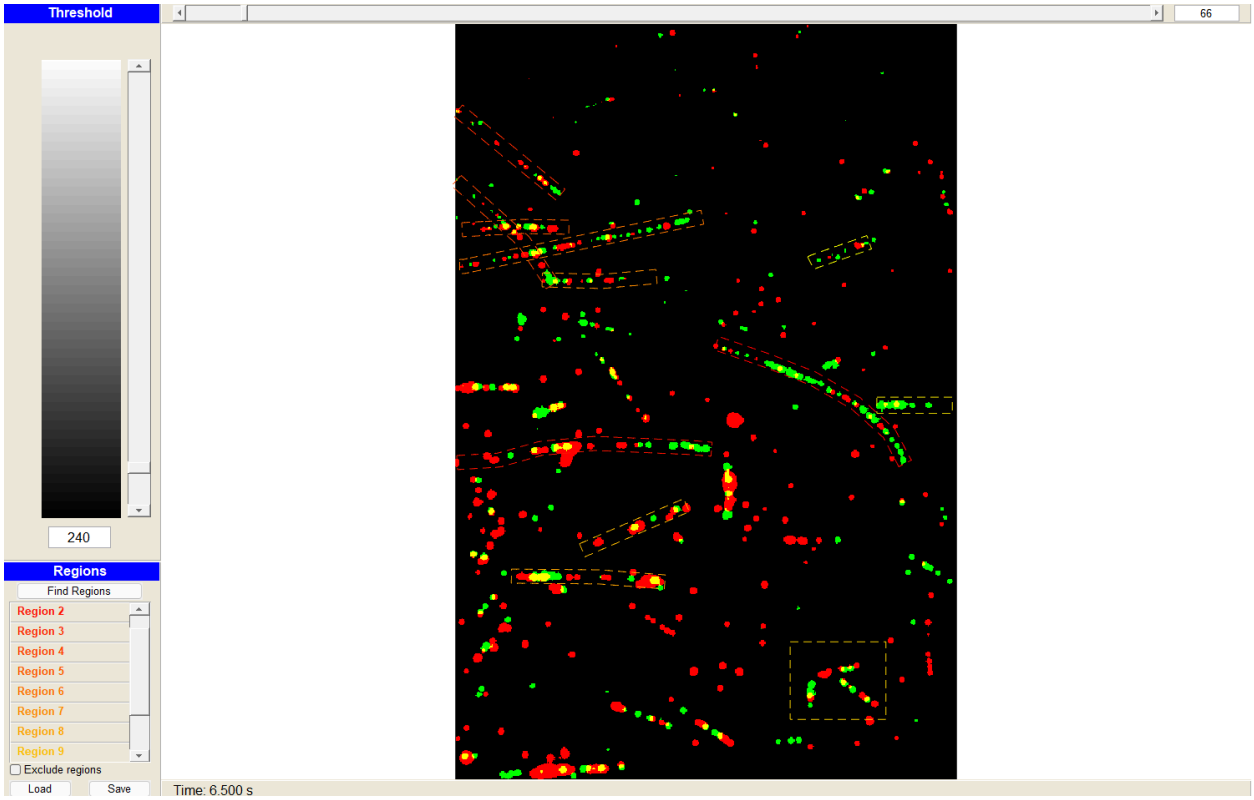


4.2. (Optional) Choosing Regions

Simply pick areas using the region selection tools outlined previously (Sec 2.2, Sec 3.1). It is not recommended to use automated region finding for stepping data in its current form.

4.3. Set Appropriate Threshold

As outlined in Sec 2.3, but make sure to do it for both channels differentially. Preferably toggle between threshold and max/min view to see a good match.



4.4. Configuration

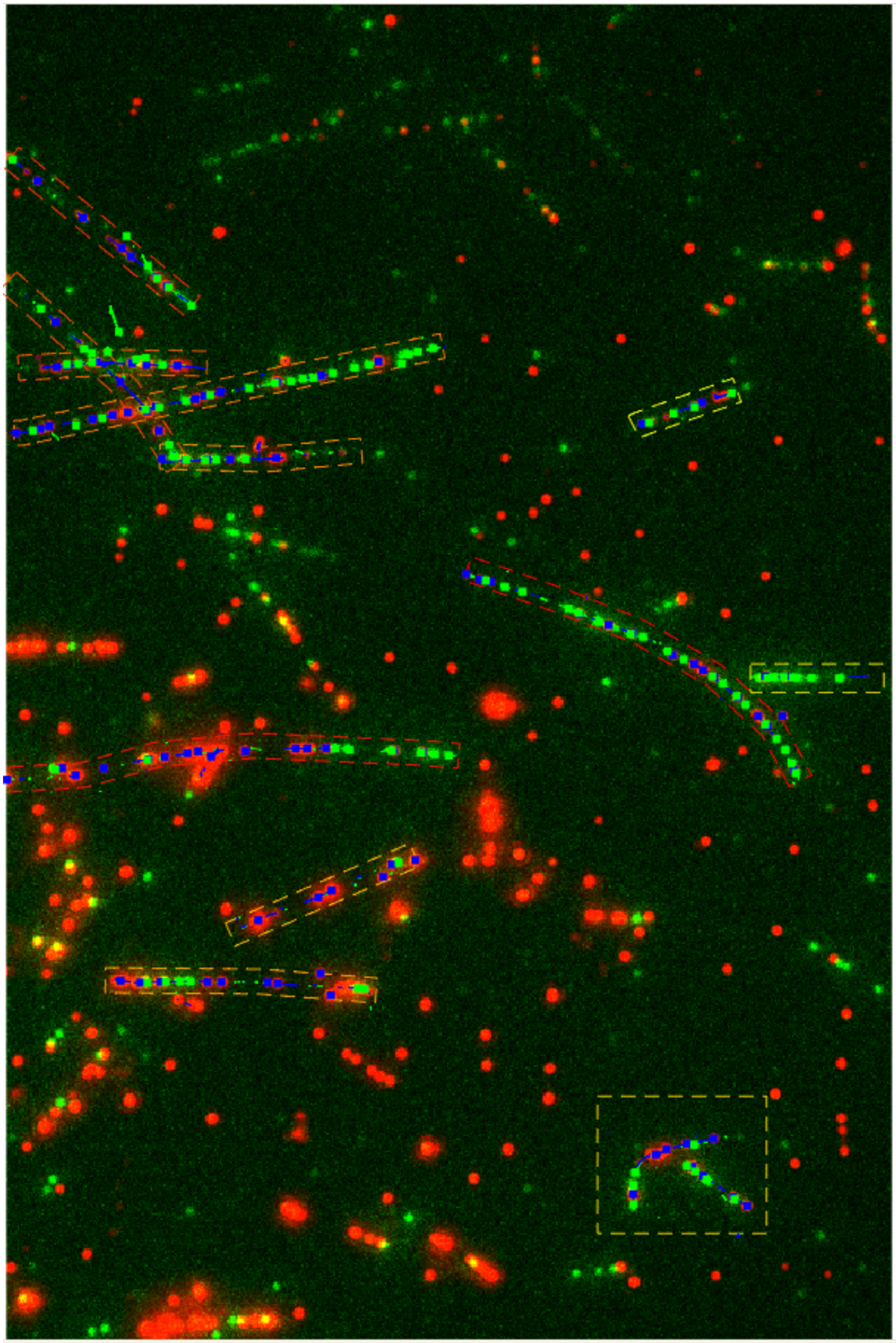
Configuration should need minimal changes if on stepping branch. High position weight is common. One thing that may be most commonly changed is the **starting frame** if in an effort to conserve dye emission you turn on the laser a bit after starting the movie. For more information on adapting configuration, check out Sec 2.4.

4.5. Add Stack(s) to Local Queue, Analyze Local Queue, Load Tracks

All of these steps follow from Sec 2.5-2.7 respectively. Make sure to add both channels separately to queue.

Analyzing stepping data can take a bit of time if not careful with region selection. There is also a high amount of post-processing due to the longer allowance for blinking and breaks. It would be recommended to start it and do something else for an hour or so and come back to it, but make sure your computer sleep mode is at a higher number of hours, or never.

When it is done, load tracks and get ready to process.



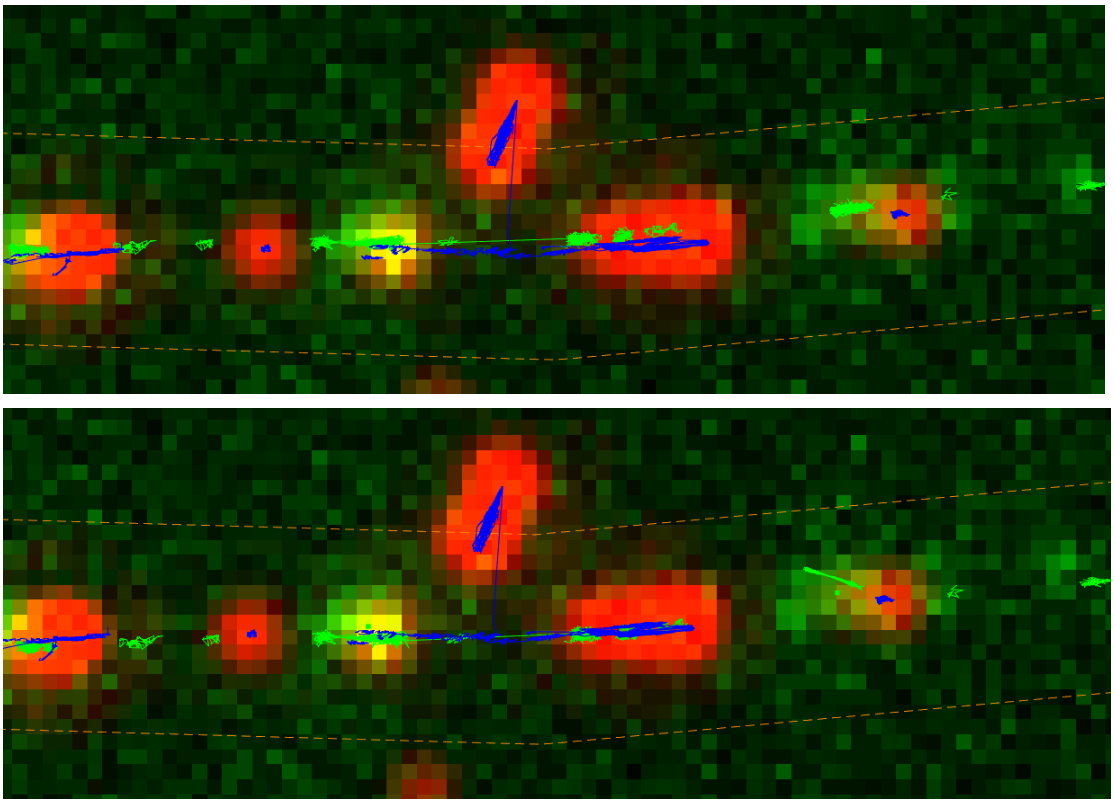
4.6. (Recommended) Toggle Multi-Channel Corrections

If the experiment requires high localization precision between channels, it may be necessary to overlay the molecule tracks. This requires a reference nanohole image or (even better) a QD fudicial image. You will need to load this in and process as a normal assay with the resulting objects stored in a FIESTA file. Information about doing a nanohole alignment assay: https://docs.google.com/document/d/11HpzRsf2ZeJU9lxOKQYcGlFjPKWdXPVXwy39fG75zbk/edit#heading=h.v9d8zs92b878

Use **Options > Set Multichannel Corrections** to make a spline interpolation of the reference image to transform the second channel onto the same coordinate system as the first. Select Channel 1's and then Channel 2's results file. Finally, once this is done, the mask will be set and you can toggle it on/off by using the **Options > Toggle Multichannel Corrections** option.

As one can see, the resolution is already pretty good, but sometimes the correction gives that extra push to be subpixel. Often, a nanohole image of about 25 images will get to subpixel resolution.

Tracks before vs tracks after multichannel corrections applied



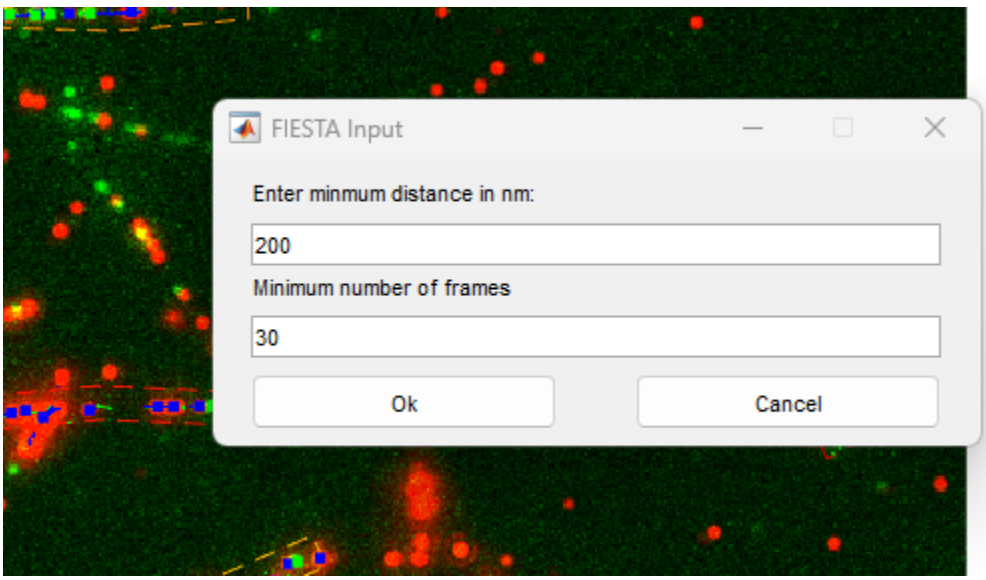
Note that the movie does not change, only the Molecule tracks do. The blue tracks are unchanged from Channel 1 (blue) and the tracks from Channel 2 (green) have been transformed to Channel 1 coordinate frame

4.7. Filter Tracks

This is where things significantly deviate from motility assays. Whereas motility assays will be looking at kymographs and mean behavior, stepping statistics require careful examination of every point to begin to draw single molecule conclusions. We have developed a pipeline to help quickly extrapolate this data from the FIESTA tracks.

As before, from the **Edit > Find all moving objects** option, select a reasonable distance and length you would like to have seen a molecule travel. This can be a very low bound.

In reality, this step is optional, but it often speeds up the next step by a substantial fraction without any data loss.

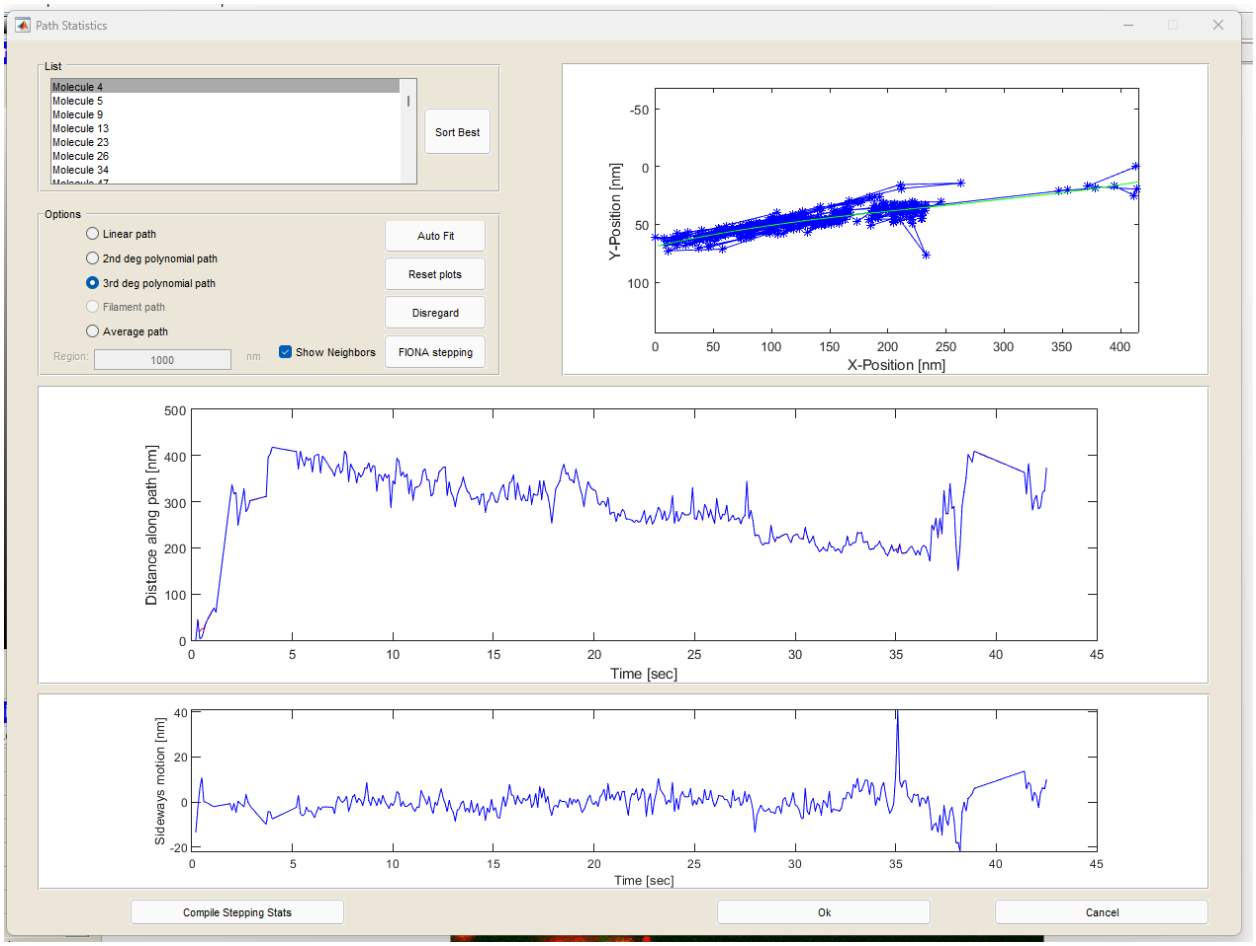


4.8. Calculate Path Statistics

Now the power of FIESTA comes into play where it will calculate a best fit path to every SELECTED molecule's trajectory. You can select a specific type of path if desired (linear, cubic, etc.) or just leave it at auto.

Simply press **Statistics > Path Statistics** and let FIESTA do it for you

At the end, a Path Statistics GUI should pop up that looks something like this

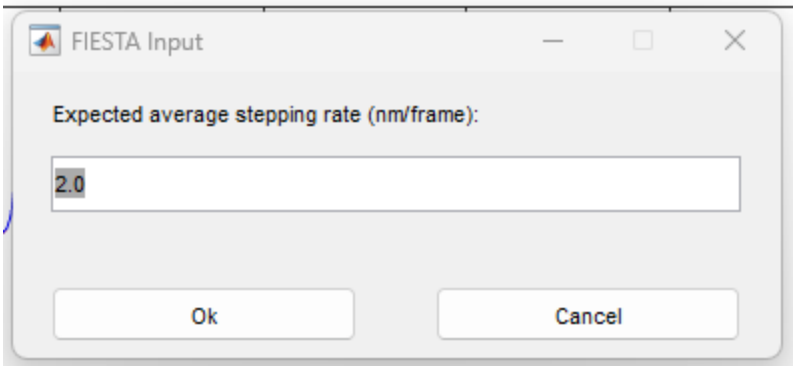


4.9. (Recommended) Sort Best

Obviously, not all of the traces are going to be great. There is a lot of noise causing agents in this sort of data. How can we grab the best molecules?

There is a custom sorting algorithm built into this PathStatsGui that will sort every trace and rank “best stepping candidates” determined by rate of movement as well as “steppiness”. More information about this can be found in the detailed architecture guide.

You will need to choose an expected average stepping rate to help guide the algorithm. In most cases, keeping it at the default **2.0 nm/frame** will not lead you astray unless your ATP conditions are much higher than stepping conditions would hope.



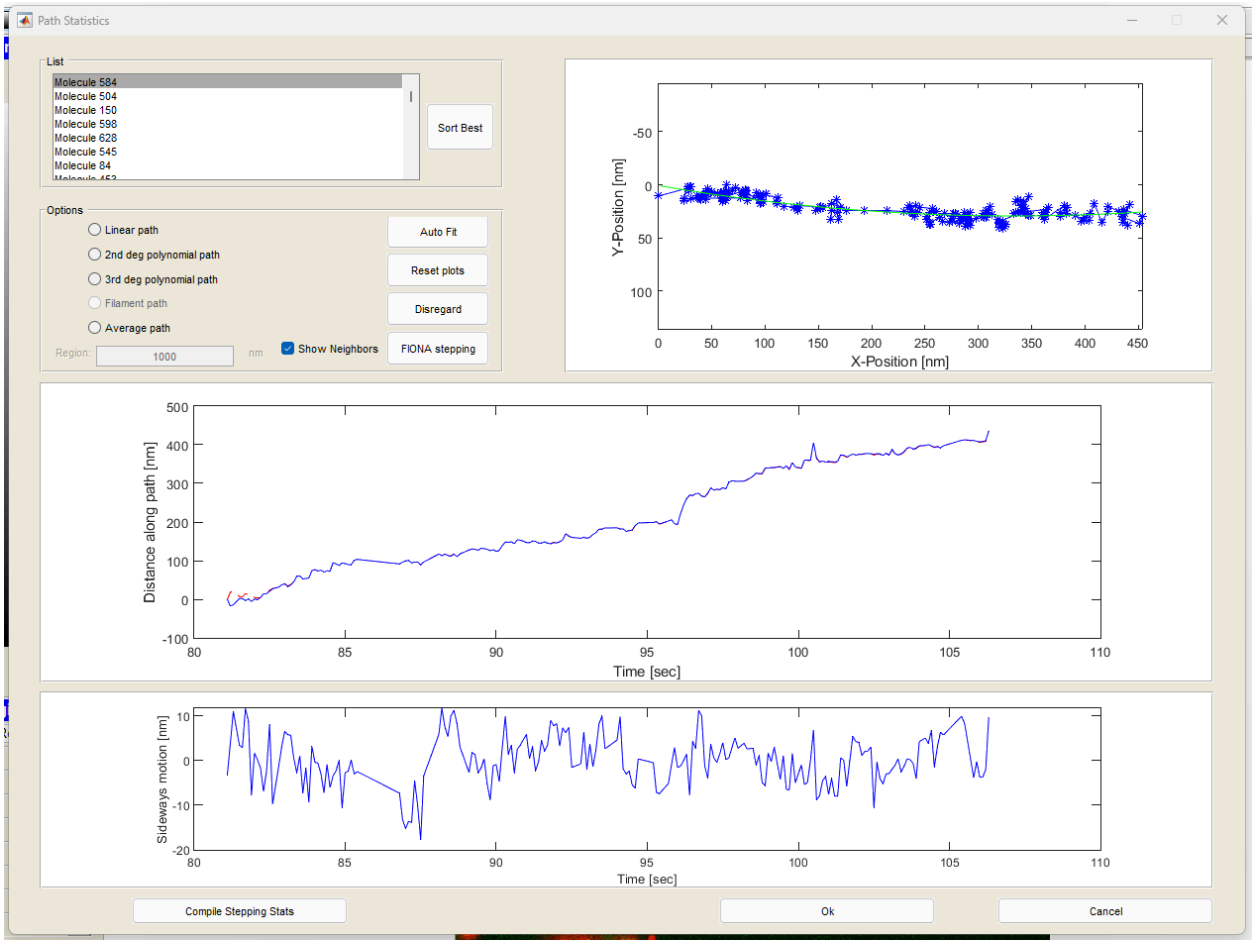
You should notice that after sorting, your first 5-10 molecules are much better than just enumerating them by increasing number. This will save you time in looking through traces, especially if you have ~100 candidates in a movie.

4.10. FIONAviewer

Now to fit steps and see data, one can use FIONAviewer, a preexisting code from before the Yildiz Lab. It uses Shwarz Information Criteria / Bayes Information Criteria to make a greatest likelihood estimate of the stepping characteristics given a trace.

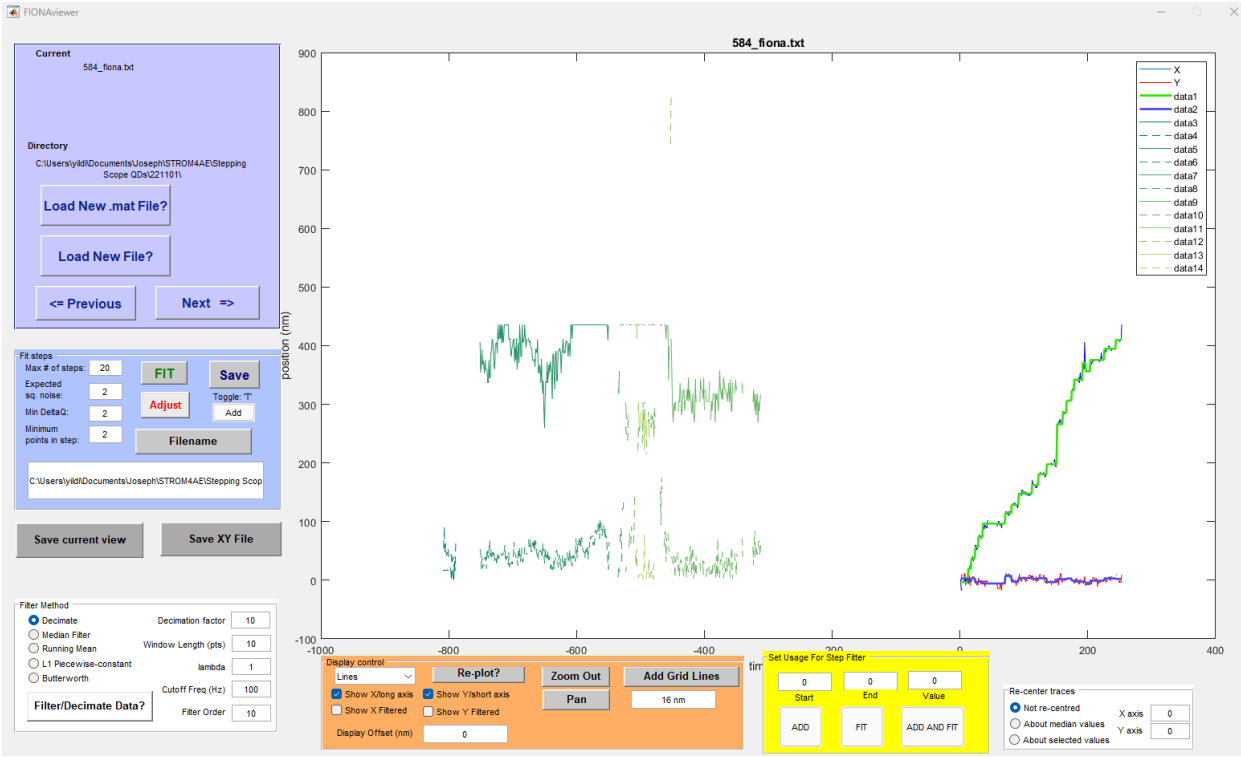
In order to access and adapt this powerful machine, once you have selected a molecule in blue, simply press the FIONAviewer button. It will automatically generate text files to be read and make a new figure plot for more user interaction.

As soon as you press FIONAviewer, the fMainGui molecule label will change to cyan. This will help you look for the molecules you analyzed on the actual image.

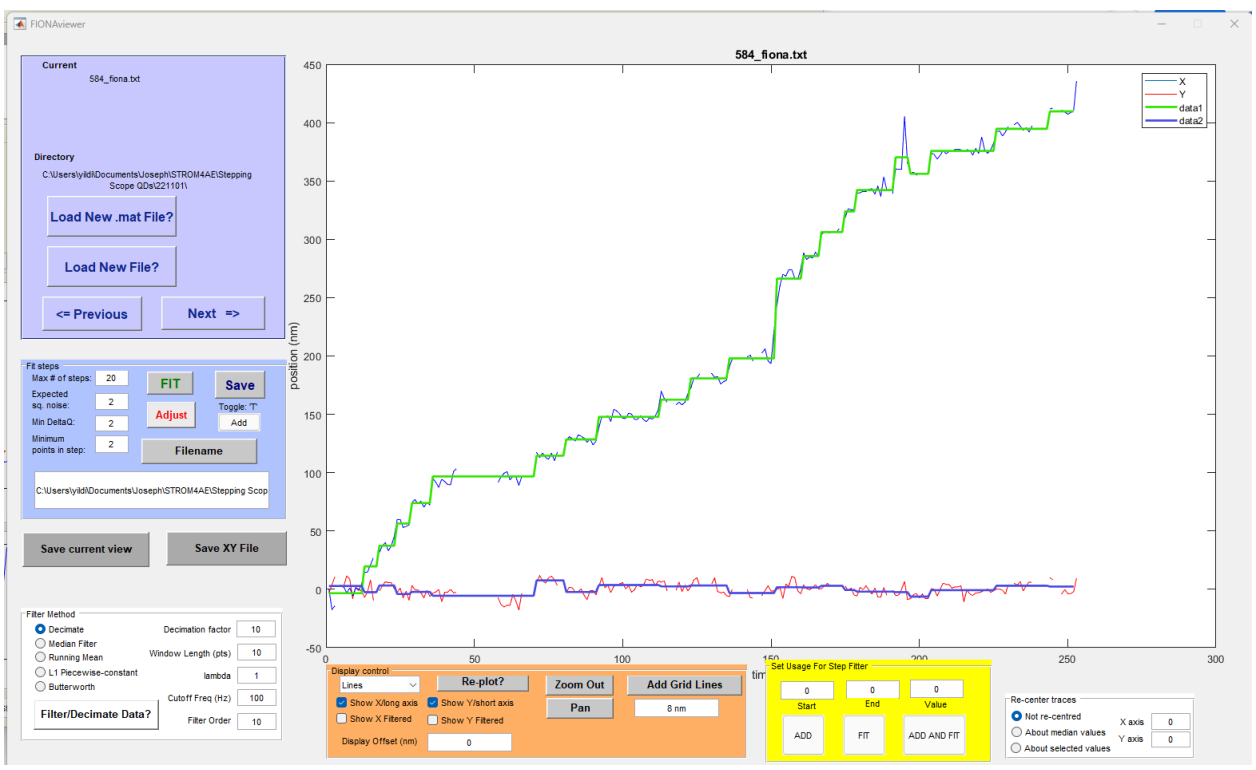


There is an extended option to **Show Neighbors**. Selecting this option will search through all molecules in the other channel and plot ones that are close to the selected molecule's trajectory. This is a useful visualization tool if you are attempting to compare molecule dynamics in the presence of a different species.

This is an example of a trace that shows neighbors where they are plotted with their longitudal (x, solid) and transverse (y, dashed) components in like colors:



And this is one without

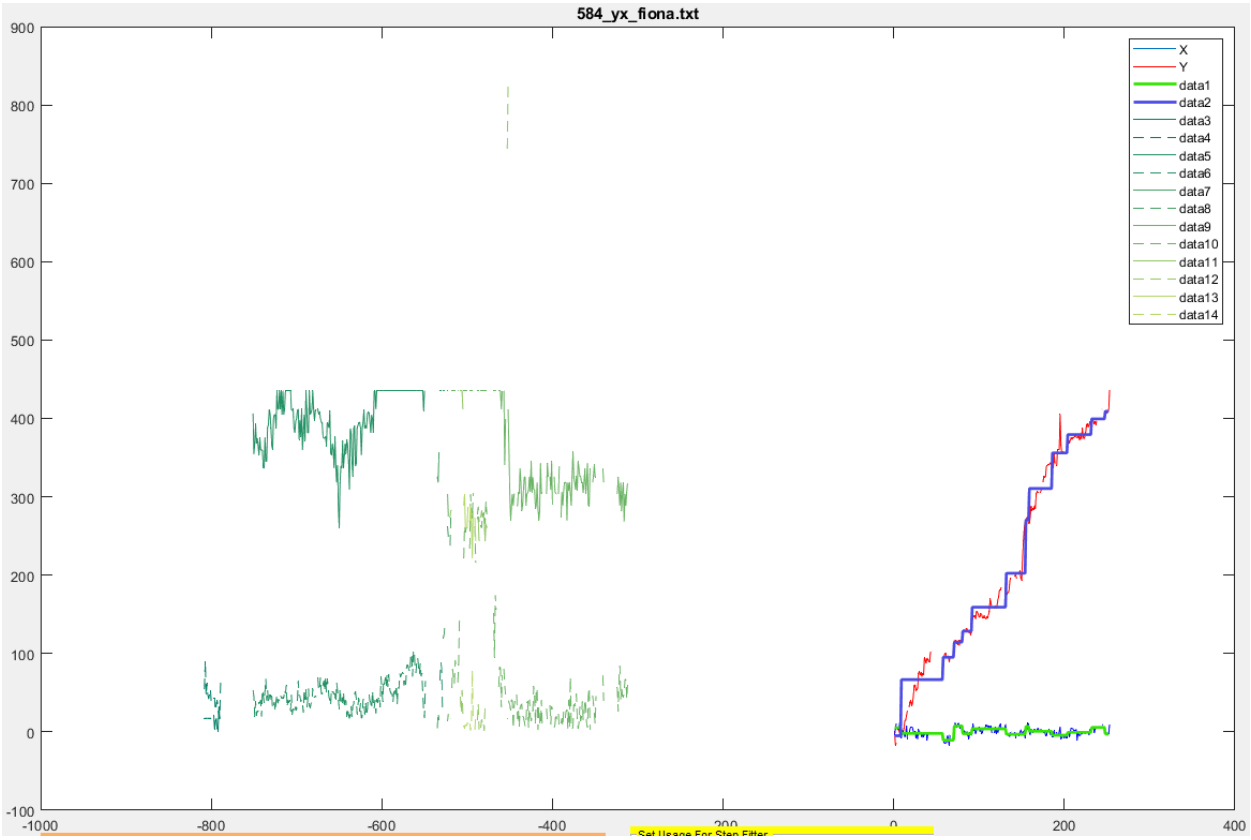


One can FIT data by selecting **FIT**. Then adjust steps either adding or deleting after pressing **adjust**. **Grid Lines** can be added at a set interval.

Save will save the mat file with step stats. This will only be the stepping stats of the selected molecule.
A figure can also be saved using **Save current view**.
The savename of these files is determined by **Filename**. However, it automatically generates to be whatever file was loaded +_xy.

If desired, to see the fit for the transverse axis, simply press next and it will switch everything to _yx. All the above steps can be repeated.

_yx fitting. Note that the main green fitting line is on the transverse axis now.



Do note that continuing to press next for further files is possible, but not wise for automatic save updating and neighbors plotting. It is recommended after completing a FIONA trace to go back to exit, go back to the path stats gui, and select the next molecule that you wish to analyze.

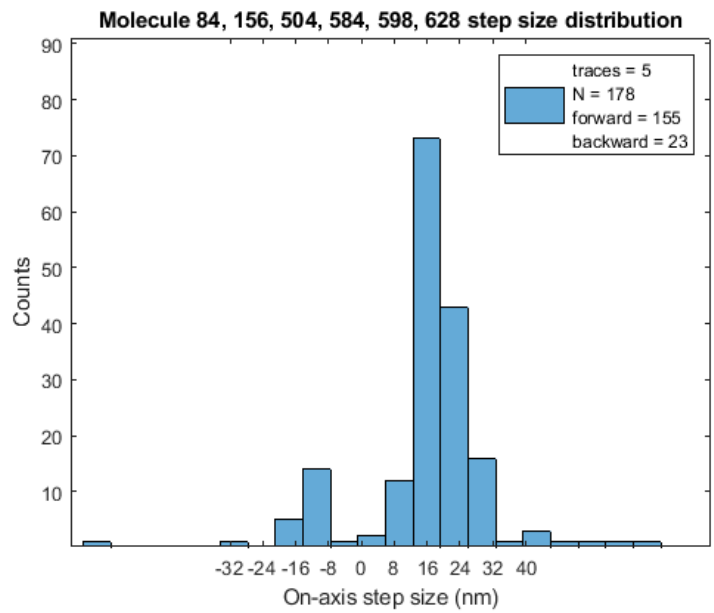
Continue this process as deemed fit

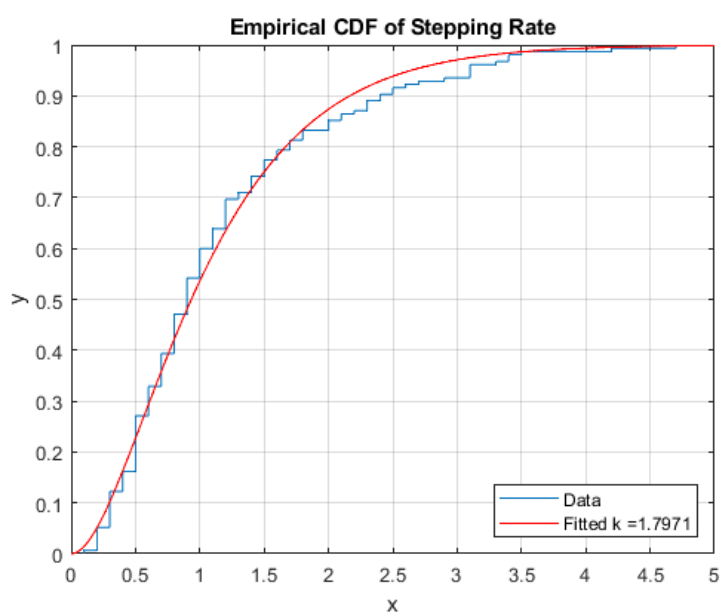
4.11. Compile Stepping Statistics

Finally, once enough data has been collected and saved, one can run composite analysis on all the xy .mat files (not .fig) to get statistics. Just make sure those

are the only type of .mat files in the folder you select. It will then return a bunch of statistics on step size, dwell times, and so on.

For some reason, you cannot touch any other GUI while in PathStats. So make sure to close this one first. It is suggested to press **Ok** to save yourself time if you want to reopen the selected molecules. Then it will save all your path statistics.





This data was taken with K490 at 200 nM ATP in the presence of MAP7. Step size was calculated to be mean 15.6 nm ± 12.3 nm with mean stepping rate of 1.79 steps/s. Could be better but not bad for only 5-10 minutes in front of the

computer. Also some noisy parts should be removed (hopefully an improvement to come)

5. Navigation

The example goes over applying a lot of different tools, but if you just want to find where they are then refer to this section. We'll go through what each section of the code does. In general it is laid out so you work left to right. Load and set tifs on the left side, analyze and select tracks on the right.

5.1. Keyboard Shortcuts

It is relatively easy to implement and change keyboard shortcuts in YFIESTA. All of this is defined in fMainGui function keypress where there is a case structure .

```
switch(key)
    case 28
        xy{l}=xy{l}-dx*0.1;
```

Every keyboard press has a "key" associated with it. Simply print key to find out what they all are. Case 28 happens to be the left arrowpad.

Due to this, there are some shortcuts that act as button presses. Here is a small list which can be adapted for user specificity if desired:

Keyboard Hotkey/Shortcut	Action
q,w,e,r	Channel 1,2,3,4 toggle respectively
t	Toggle on/off threshold view
y	Add current channel to local queue
a	Analyze local queue

5.2. Main Toolbar

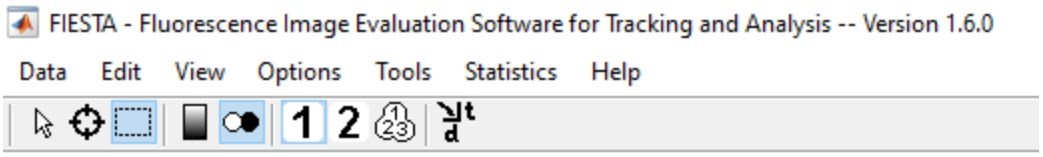


Image 3.1 - Basic Toolbar for Multi-Channel/Single-Channel Acquisition

Data - load stacks of images, load processed tracks, load objects, and also remove them

Edit - Edit loaded data or stacks. This includes adding stacks to queue, filtering molecules, and joining tracks together

View - Do basic ImageJ processes such as Z-project, apply corrections, color overlay, and export the current view for reference

Options - Apply configurations previously saved (such as intensity vals, regions, etc) stored in the processed data array under Config. This is particularly useful for replicating data or keeping params the same across movies.

Tools - Do basic measurements using geometric objects

Statistics - Contains the options to do the bread and butter of analysis on loaded tracks. Path statistics, velocity, motility, MSD, Kymograph, basically most things. It will only apply it to selected tracks. (highlighted blue in the Data tab on the right side)

Help - redirects to basic information on FIESTA

Gray-Scale Bar

Region Selector Toolbar (Mouse Pointer, Scope Tracker, Dotted Rectangle) - Select desired (or undesired) molecules and regions. Mouse selects one, Scope tracker traces out a free-hand region, Dotted rectangle selects a rectangular area. After selecting a region it shows up in the left sidebar Regions section

View Mode (Grayscale colorbar, white and black circle) - Toggle between brightness-contrast view and threshold view. Note that threshold view is what is set according to the analysis to determine peaks. *Hotkey: t*

(Only in multichannel)
Channel View (1, 2, …, venn diagram) - Use 1 and 2 to select which molecules and stacks you see. Use the venn diagram to view both channels simultaneously overlayed and color coded. *Hotkey: q,w,e,r*

5.3. Left Sidebar

5.4. Right Sidebar

6.